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Robust drug recommendation based on patient status awareness and unbiased prediction

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ABSTRACT

The AI-based drug recommendation system can recommend a suitable series of drugs for patients, which is a significant application in current healthcare field. However, current drug recommendation methods overlook personalized differences among patients, leading to the spread of personalized information among the generic patient populations, thereby resulting in information bias. Additionally, patients' historical visits are not always strongly correlated, indicating that the patient may not always suffer from same or similar diseases in different visits. Therefore, we propose Robust Drug Recommendation Based on **P**atient Status **A**wareness and **U**nbiased **P**rediction (PAUP). First, we introduce a concept of the "Standardized Doctor" module, which constrains the input and output of the model to capture patient-agnostic generic treatment strategies, thereby avoiding information bias caused by personalized patient differences. Then, we design a personalized patient information mining module that explicitly captures patient-specific information by mapping biases between generic and real drug representations. Through the two aforementioned modules, we formulates personalized treatment strategies for patients. Finally, we employ attention mechanisms to attenuate the influence of weakly correlated historical visits on current visit, which is utilized as auxiliary information to guide the final recommendation results. Additionally, we model both 2D and 3D molecular structures of drugs to preprocess drug representation. Experimental results on MIMIC-III and MIMIC-IV datasets demonstrate that PAUP outperforms the state-of-the-art in drug recommendation. In addition, compared to the suboptimal baseline, PAUP saves 59% of training time and 38% of inference time, with a faster convergence rate. The code is available at https://anonymous.4open.science/r/PAUP-master.

1. Introduction

Drug recommendation extracts patient visit information, such as diagnose, procedure, and drug records, from comprehensive longitudinal electronic health records (EHRs) to recommend appropriate drug combinations for patient treatment. It aids doctors and researchers in making precise and efficient medical decisions, which obtains significant attention in recent years.

Traditional drug recommendation models rely on instance-based methods (Gong et al., 2021; Wang et al., 2019; Zhang et al., 2017), which consider only the clinical data from the patient's current visit (e.g., diagnose and procedure records). These methods

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neglect the patient's historical visit information, which may lead to the misjudgment of the patient's condition (e.g., acute or chronic), thereby significantly impact the treatment outcomes. To address this issue, numerous longitudinal-based methods have been proposed. These methods take into account both the current and historical visit records of the patient. By constructing temporal models, they learn knowledge from the patient's historical visits, enabling more accurate drug recommendations based on the clinical data from the patient's current visit. Typically, in the patient visit process, diagnose and procedure records are treated equally (Choi et al., 2016; Shang et al., 2019a,b; Sun et al., 2022; Yang et al., 2021a), which are simply concatenated or summed to represent the clinical data for the visit. Considering the potential associations among different clinical data, some methods further construct bipartite relational network (Li et al., 2024a), Transformer (Zhao et al., 2024b), or cross-attention (Tang et al., 2024) to achieve clinical data fusion. Additionally, to explore the physicochemical properties of drugs, certain methods (Bhoi et al., 2023; Yang et al., 2021b, 2023b) incorporate 2D molecular structures (including SMILES (Simplified Molecular Input Line Entry System) sequences or atomic connectivity networks) to model atom-level drug representations. Building on this, more knowledge such as hypergraphs (Wu et al., 2023b), drug motifs (Zhao et al., 2024a), and 3D molecular structures (Mu et al., 2025) are introduced to further explore and model the complex properties of drugs. Moreover, considering that a patient's historical visit records are not always strongly correlated, some studies retain only similar historical records (Kim et al., 2024; Wang et al., 2021), while others incorporate highly similar visit records from other patients (Wu et al., 2023a, 2024) to improve the accuracy of drug recommendations.

However, current longitudinal-based drug recommendation methods overlook the capture of personalized patient status information. Although previous approaches could implicitly derive personalized patient status information from patients' historical visit data, this information is highly entangled with their treatment information, leading to suboptimal drug recommendation performance. Our approach extracts patient-agnostic generic drug predictions (referred to as the "Standardized Doctor") from patient visit data, and explicitly captures personalized patient status information by evaluating the bias between these generic and the real drug prescriptions. This strategy effectively mitigates the interference of biased information, thereby enhancing the robustness of the model. Furthermore, we also consider the potential changes in personalized patient status information across different visits, which aligns more closely with real patient status.

In summary, we propose Robust Drug Recommendation Based on **P**atient Status **A**wareness and **U**nbiased **P**rediction (PAUP). First, we construct the "Standardized Doctor," which derives patient- agnostic generic treatment strategies from clinical data. Then, we explicitly identify and extract personalized state information of patients, and further capture the variations of this information across patient visits. Finally, we utilize drug data from patients' historical visits as auxiliary information to guide the ultimate drug recommendation outcomes. Additionally, during the preprocessing stage, we incorporate the 2D and 3D molecular structures of drugs to analyze their physicochemical properties and introduce drug-drug interaction (DDI) constraints to reduce the risk of potential adverse effects. The experimental results show that on both the MIMIC III and MIMIC IV datasets, PAUP outperforms the state-of-the-art (SOTA) in drug recommendation. Furthermore, compared to the main baselines, PAUP exhibits a significant advantage in efficiency.

The primary contributions of the paper are as follows:

1) We are the first to propose the concept of the "Standardized Doctor" in drug recommendation. This "doctor" prescribes generic-purpose drugs for treatment solely based on clinical data, which are independent of specific patients. This implies that it is not influenced by the information bias caused by patient-specific variability.
2) We explicitly identify and extract the personalized status information of patients in drug recommendation, which explains the discrepancy between the generic drugs prescribed by the "Standardized Doctor" and the real drugs prescribed in practice. Furthermore, we track the variations of historical status information across patient visits to better understand the patient's current status.
3) We propose an auxiliary drug recommendation method that captures the importance between the currently predicted drugs and the historically prescribed drugs. This method leverages historical drug information to assist and guide current drug recommendations.
4) We design a preprocessing process for drug representation, which explores the physicochemical properties of drugs by capturing their 2D and 3D molecular structures, thereby achieving high-quality drug representations.

The remaining sections of the paper are organized as follows. Section 2 is the related work. Section 3 provides a detailed description of the proposed PAUP model. Section 4 is the experiments, and Section 5 concludes the paper.

2. Related work

2.1. Recommendation systems

Recommendation systems aim to suggest suitable items to users based on existing information. LightGCN (He et al., 2020) employs a simplified graph convolutional network (GCN) to capture user-item interaction information, effectively uncovering users' real preferences. GCN achieves message passing within the network through neighborhood information aggregation and node updates. Building on this, ApeGNN (Zhang et al., 2023) adaptively adjusts the aggregation weights of nodes based on local node information. SSNet (Song et al., 2022) introduces a learning mechanism for node representations during the aggregation process, addressing the issue of scale distortion inherent in aggregation. IHGCL (Sang et al., 2025) captures fine-grained information by exploring the latent intentions embedded in user-item meta-paths, while Coase (Li et al., 2025a) captures fine-grained dynamic user preferences by modeling the local and global contexts to capture local and global contexts. FICLRec (Su et al., 2025) utilizes frequency information from users' latent intentions to improve the recognition of high-frequency intent. Moreover, SGL (Wu et al., 2021) and XSimGCL (Yu

et al., 2023) incorporate contrastive learning to enhance the robustness of recommendation systems. Both methods construct new views for contrastive learning by artificially perturbing the original views. To improve the genericization capability of contrastive learning, AutoCF (Xia et al., 2023) and AdaGCL (Jiang et al., 2023) utilize graph autoencoders to adaptively reconstruct new views, while ADGA (Ma et al., 2024a) leverages a memory-based graph attention mechanism to adaptively perturb the original views. KCLM (Chen et al., 2024) constructs two contrastive learning tasks to respectively address the differences in long-term and short-term interests and model the commonalities between clicked products and corresponding keywords. SCL (Chen & Zhu, 2025) designs two data augmentation mechanisms to capture visiting uncertainties. Furthermore, to mitigate the sparsity of recommendation systems, CLDS (Ma et al., 2024b) and HHGSA (Khan et al., 2025) incorporate social network information to enrich user connection relationships within the network. The former leverages Gaussian distribution, pre-training, and contrastive learning to enhance the robustness of the model, while the latter captures preference similarities between friends and strangers to model higher-order relationships in the network. Moreover, GCMR (Ma et al., 2025) incorporates multi-behavioral user information to enhance the interactions between users and items within the network.

2.2. Drug recommendation

In comparison to traditional recommendation systems, drug recommendation poses greater challenges in data processing and modeling. In the process of drug recommendation, models need to consider additional factors such as potential medication reactions or withdrawals resulting from different drug-drug interaction (DDI). Based on a thorough consideration of these factors, drug recommendation can be categorized into two main types: Instance-based and Longitudinal-based. Instance-based models solely consider the clinical data from the patient's current visit. Compnet (Wang et al., 2019) constructs a dynamic drug knowledge graph to assist in clinical treatment. Leap (Zhang et al., 2017) incorporates external clinical knowledge into reward mechanisms and adaptively determines the required number of drugs. SMR (Gong et al., 2021) bridges electronic medical records and medical knowledge graphs to construct high-quality heterogeneous networks for drug prediction. However, the aforementioned methods overlook the patient's medical history, which may compromise treatment efficacy for specific individuals and even introduce potential safety risks. Consequently, longitudinal-based models that incorporate historical patient visits become mainstream. Leveraging the sequential nature of patient visits, MICRON (Yang et al., 2021a) employs cascaded residual networks to capture historical changes in patient visits, while SafeDrug (Yang et al., 2021b) and DPID (Li et al., 2025b) utilize RNN/Bi-GRU architectures to selectively retain long-term visit information. In contrast, LamRec (Tang et al., 2024) and Trans-GAHNet (Li et al., 2024c) adopt Transformer mechanisms to weigh the importance across different historical visits. To balance model complexity, LMGA (Fan et al., 2024) proposes a lightweight drug recommendation framework enhanced with a memory module to capture localized historical visit information. To enhance drug representation, SafeDrug (Yang et al., 2021b), Molerec (Yang et al., 2023b), and BiMoRec (Mu et al., 2025) introduce drug molecular structure information. Nevertheless, these longitudinal-based models primarily focus on holistic modeling of patient visits without explicitly disentangling different types of information. Cognet (Wu et al., 2022), SSPNet (Zhang et al., 2025), and CT-PASMR (Ge et al., 2025) separate historical visit information from current visit data, which jointly predicting drug combinations through distinct modeling of the probabilities for generating new drugs and copying historical medications. In contrast, our proposed PAUP model separates the treatment strategy within each visit into a generic treatment component and a personalized adjustment component. This separation mitigates model bias stemming from patient variations and constructs a novel paradigm for drug recommendation.

3. Methodology

We propose Robust Drug Recommendation Based on **P**atient Status **A**wareness and **U**nbiased **P**rediction (PAUP). The method, for the first time, proposes the construction of a generic and patient-agnostic drug prediction model in drug recommendation, and explicitly explores and tracks personalized patient status information. Then, by leveraging preprocessing techniques, the method investigates the physicochemical properties of drugs, which is used to assist in guiding drug recommendations under the current clinical treatment outcomes.

3.1. Problem formulation

Let $\mathscr{R} = \{r^1, \dots, r^N\}$ denote the set of all patients, where N represents the total number of patients. In paper, we primarily extract patients' clinical records $\mathscr{X} = \{\mathscr{D}, \mathscr{P}\}$ and prescribed drug records $\mathscr{M} = \{m^1, \dots, m^N\}$ from longitudinal electronic health records (EHR), where $\mathscr{D} = \{d^1, \dots, d^N\}$ represents the set of all diagnosis records of the patients, and $\mathscr{P} = \{p^1, \dots, p^N\}$ is the set of all procedure records. Furthermore, the set of diagnosis records for the k-th patient across multiple visits is denoted as $d^k = \{d^k_1, \dots d^k_{T_k}\}$, the set of procedure records as $p^k = \{p^k_1, \dots p^k_{T_k}\}$, and the set of drug record as $m^k = \{m^k_1, \dots, m^k_{T_k}\}$, where T_k represents the total number of visits. Let D, P, and M represent the sets of all possible diagnoses, procedures, and drugs, respectively. For the t-th visit of the k-th patient, the corresponding diagnosis records are denoted as $d^k_t = \{d^k_{t,1}, \dots, d^k_{t,\Gamma^k_{t,d}}\}$, the procedure records as $p^k_t = \{p^k_{t,1}, \dots, p^k_{t,\Gamma^k_{t,p}}\}$, and the drug records as $m^k_t = \{m^k_{t,1}, \dots, m^k_{t,\Gamma^k_{t,m}}\}$, where $\Gamma^k_{t,d}, \Gamma^k_{t,p}$ and $\Gamma^k_{t,m}$ denote the number of diagnosis records, procedure records and drug records, respectively, $d^k_{t,*} \in D$, $p^k_{t,*} \in P$, $m^k_{t,*} \in M$. Let $A \in \mathbb{R}^{|M|\times|M|}$ denote the drug-drug interaction (DDI) matrix, where $A[i,j] = 1$ indicates a harmful interaction between drug i and drug j, and $A[i,j] = 0$ otherwise.

The objective of the drug recommendation task is to predict the probability $y^k \in [0,1]^{|M|}$ of each drug being recommended for a given patient k. This probability is determined by comprehensively considering the patient's diagnosis record set d^k, procedure

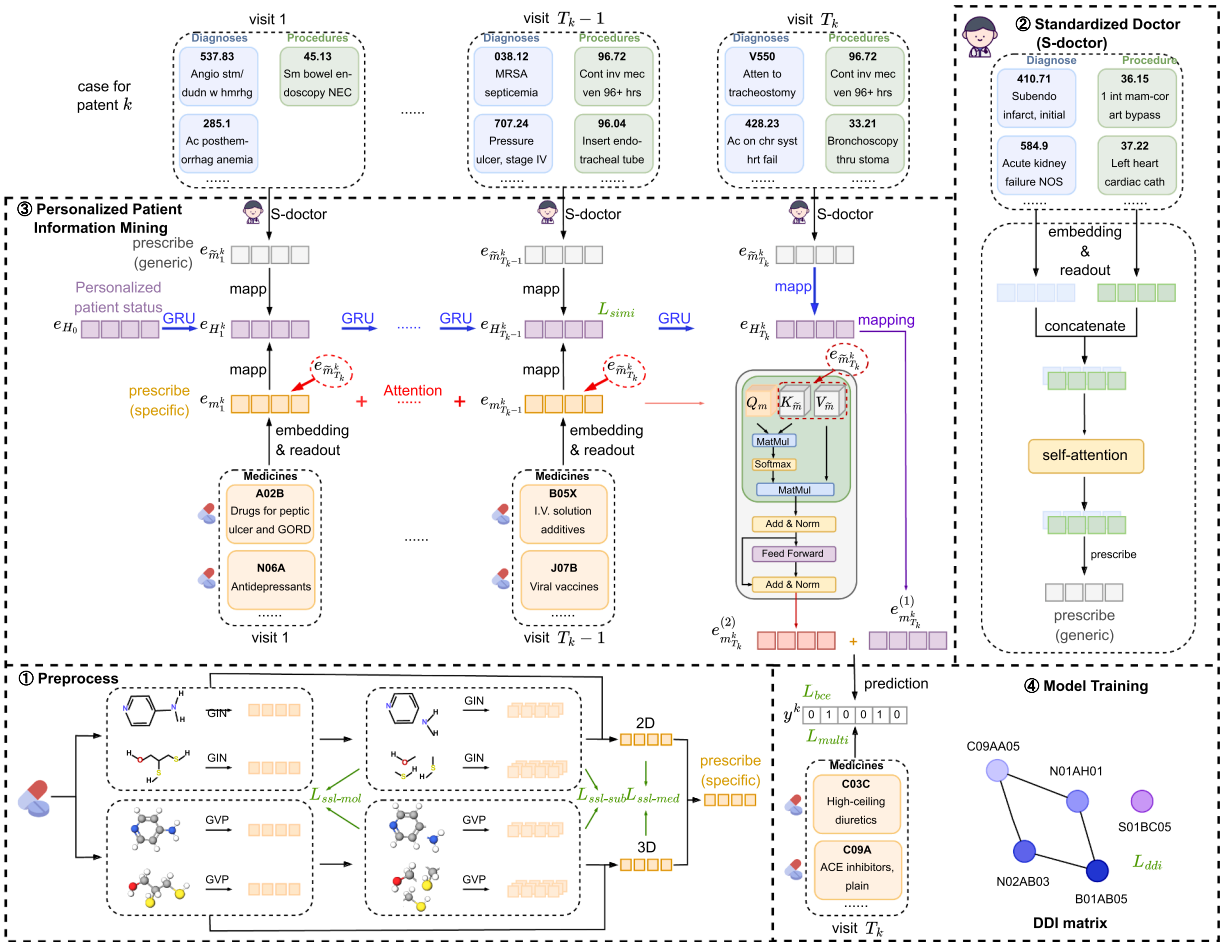


Fig. 1. Algorithm Framework, where the Preprocess module explores 2D and 3D molecular information of drugs to obtain representation of drugs containing personalized patient status information as prescribed by the Drug Recommendation module, the S-doctor module uncovers generic diagnosis and procedure information to obtain the generic drugs representation of Drug Recommendation module. In Drug Recommendation module, the blue arrows represent the main pathway for predicting drug representation, while the red arrows indicate auxiliary pathways to enhance the stability of drug representation. Additionally, we consider potential risks resulting from drug-drug interactions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

record set p^k, historical drug record set $m^k_{1:T_k-1}$, DDI matrix A, and the visit records of other patients. The predicted probabilities are expected to align with the real prescribed drug records $m^k_{T_k}$.

3.2. Algorithm framework

As shown in Fig. 1, the algorithm framework of PAUP is mainly composed of four components.

1) Preprocess Module: We incorporate both 2D and 3D molecular structural information of drugs to capture their physicochemical properties, thereby improving the quality of drug representations.
2) Standardized Doctor Module: We construct the Standardized Doctor (S-doctor), which prescribes generic drugs solely based on clinical data without considering individual patient status. This enables unbiased prediction independent of patient-specific factors.
3) Personalized Patient Information Mining Module: We extract personalized patient status information and track the variations in patient status across different visits.
4) Model Training Module: We incorporate the DDI matrix and historical drug prescription records, among other information, to inform and guide the drug recommendation process.

3.3. Preprocess module

We capture the physicochemical properties of drugs from the 2D and 3D structures of the molecules, and further learn drug functions from the substructures of the molecules. Molecules are composed of atoms bound together by covalent bonds. Inspired by

BiMoRec, we use the RDKit library[1] to extract the chemical properties and structural features of atoms, which include atomic number, aromaticity, hybridization state, chirality tag, and degree. For the given drug x, the model uses GIN (Xu et al., 2018) to implement neighborhood aggregation and updating of atomic representations in the molecule, and then computes the mean of atomic representations to obtain the 2D molecular representation set $e^x_{2D-mol} = \{e^x_{2D-mol,1}, e^x_{2D-mol,2}, \cdots, e^x_{2D-mol,T_x}\} \in \mathbb{R}^{T_x \times dim}$, where T_x represents the number of molecules in drug x, $e^x_{2D-mol,1}$ denotes the representation of the 1-st molecule in drug x. Construct a 3D molecular network by treating atoms as nodes and connections as edges if the distance between atoms is less than 4.5$\mathring{A}$ (Townshend et al., 2021). Subsequently, use GVP (Jing et al., 2020) to directly extract 3D molecular representations $e^x_{3D-mol} = \{e^x_{3D-mol,1}, e^x_{3D-mol,2}, \cdots, e^x_{3D-mol,T_x}\} \in \mathbb{R}^{T_x \times dim}$. This method eliminates the need for traditional data augmentation through rotation and translation, thereby improving both performance and computational efficiency. The properties of a molecule are often determined by specific substructures. Using the BRICS (Yang et al., 2023b) method, 2D molecular substructures are obtained. Then, the GIN is employed to extract substructure representations $e^x_{2D-sub} = \{e^x_{2D-sub,1}, e^x_{2D-sub,2}, \cdots, e^x_{2D-sub,\Gamma_x}\} \in \mathbb{R}^{\Gamma_x \times dim}$, where Γ_x is the number of substructures in drug x. Subsequently, 3D molecular substructures are constructed using the API provided by RDKit, and then GVP is again utilized to extract 3D molecular substructure representations $e^x_{3D-sub} = \{e^x_{3D-sub,1}, e^x_{3D-sub,2}, \cdots, e^x_{3D-sub,\Gamma_x}\} \in \mathbb{R}^{\Gamma_x \times dim}$.

Next, we employ cross-attention to obtain the 2D and 3D representations of the drug. Take 2D as an example, the query matrix $Q_{mol} \in \mathbb{R}^{1 \times dim}$, key matrix $K_{sub} \in \mathbb{R}^{1 \times dim}$ and value matrix $V_{sub} \in \mathbb{R}^{1 \times dim}$ are constructed as

$$Q_{mol} = mean(e^x_{2D-mol})W^Q_1, K_{sub} = mean(e^x_{2D-sub})W^K_1, V_{sub} = mean(e^x_{2D-sub})W^V_1 \tag{1}$$

where W^Q_1, W^K_1 and W^V_1 are the weight matrices. Then, the 2D drug representation $e^x_{2D-m} \in \mathbb{R}^{1 \times dim}$ is

$$e^x_{2D-m} = softmax(\frac{Q_{mol}K^{Tr}_{sub}}{\sqrt{dim}})V_{sub} \tag{2}$$

where $\cdot^{Tr}$ denotes the transpose of a matrix or vector, and dim represents the embedding dimension. Similarly, the 3D drug representation is $e^x_{3D-m} \in \mathbb{R}^{1 \times dim}$.

We align molecular representation, substructure representation, and drug representation separately to achieve semantic and structural matching between 2D and 3D representations, thereby enhancing the robustness of the representations. Specifically, the molecular representation loss is defined as

$$L_{ssl-mol} = -\frac{1}{\Gamma}\sum_{i=1}^{\Gamma}\frac{exp(e_{2D-mol,i} \cdot e_{3D-mol,i})/\tau}{\sum_{j \neq i}^{\Gamma} exp(e_{2D-mol,i} \cdot e_{3D-mol,j})/\tau} \tag{3}$$

where Γ represents the number of total molecules, $e_{2D-mol,i}$ denotes the representation of the i-th molecule, τ is the temperature coefficient. Similarly, the substructure representation loss $L_{ssl-sub}$ and the drug representation loss $L_{ssl-med}$ can be obtained. Consequently, the final preprocess loss for drug x is

$$L_{pro} = L_{ssl-mol} + L_{ssl-sub} + L_{ssl-med} \tag{4}$$

The final high-quality drug representation set is denoted as $e_m = \{e^1_m, \cdots, e^{|M|}_m\} \in \mathbb{R}^{|M| \times dim}$, where

$$e^x_m = (e^x_{2D-m} + e^x_{3D-m})/2 \tag{5}$$

, and $x \in \{1, 2, \cdots, |M|\}$.

We use *pre-emb* to represent the entire preprocess module, and the above process can be expressed as

$$e^x_m = pre\text{-}emb(x) \tag{6}$$

3.4. Standardized doctor module

We propose the “Standardized Doctor” (S-doctor), a "doctor" that prescribes patient-agnostic generic drugs based on clinical information, which achieves the transmission of unbiased and shared generic treatment strategies across different patients. Specifically, in “Standardized Doctor”, we capture patient-specific information and constrain it from patient historical visits records with a loss function. Subsequently, by eliminating the influence of the specific patient information from the patient-specific treatment strategy, we obtain generic treatment strategy outputs.

Let $d = \{d_1, \cdots, d_{\Gamma_d}\} \in \mathbb{R}^{\Gamma_d}$ and $p = \{p_1, \cdots, p_{\Gamma_p}\} \in \mathbb{R}^{\Gamma_p}$ represent the diagnosis and procedure records set in the clinical information, where Γ_d and Γ_p denote the number of diagnosis and procedure records, respectively. The corresponding representations of the diagnosis and procedure records set are then denoted as $e_d = \{e_{d_1}, \cdots, e_{d_{\Gamma_d}}\} \in \mathbb{R}^{\Gamma_d \times dim}$ and $e_p = \{e_{p_1}, \cdots, e_{p_{\Gamma_p}}\} \in \mathbb{R}^{\Gamma_p \times dim}$.

First, we process the diagnosis and procedure sets separately to obtain robust representation set. Taking diagnosis as an example, the diagnosis representation $e_{\tilde{d}} \in \mathbb{R}^{1 \times dim}$ is

$$e_{\tilde{d}} = readout(drop(e_d)) \tag{7}$$

[1] https://www.rdkit.org

where $drop$ means dropout function, and readout uses vector summation by default. Similarly, we get the procedure representation $e_{\widetilde{p}} \in \mathbb{R}^{1\times dim}$.

Subsequently, a Transformer is employed to thoroughly capture the interaction information between diagnosis and procedure representations. By concatenating diagnosis and procedure representation $e_{\widetilde{f}} = cat(e_{\widetilde{d}}, e_{\widetilde{p}}) \in \mathbb{R}^{2\times dim}$, we construct query matrix $Q_{\widetilde{f}} \in \mathbb{R}^{2\times dim}$, key matrix $K_{\widetilde{f}} \in \mathbb{R}^{2\times dim}$ and value matrix $V_{\widetilde{f}} \in \mathbb{R}^{2\times dim}$ as

$$Q_{\widetilde{f}} = e_{\widetilde{f}} W_2^Q, K_{\widetilde{f}} = e_{\widetilde{f}} W_2^K, V_{\widetilde{f}} = e_{\widetilde{f}} W_2^V \tag{8}$$

where W_2^Q, W_2^K and W_2^V are weight matrices. Then the representation $e_{\widetilde{f}'} \in \mathbb{R}^{2\times dim}$ after attention is

$$e_{\widetilde{f}'} = LN(Q_{\widetilde{f}} + softmax(\frac{Q_{\widetilde{f}} K_{\widetilde{f}}^{Tr}}{\sqrt{dim}}) V_{\widetilde{f}}) \tag{9}$$

where LN is layernorm. The representation is processed through a two-layer Feed-Forward Networks (FFN) as

$$e_{\widetilde{f}''} = LN(e_{\widetilde{f}'} + FFN(e_{\widetilde{f}'})) \tag{10}$$

where $e_{\widetilde{f}''} \in \mathbb{R}^{2\times dim}$.

Finally, we concatenate the diagnosis and procedure representations $e_{\widetilde{f}^\circ} = cat(e_{\widetilde{f}''}) \in \mathbb{R}^{1\times 2dim}$. Then, through neural network mapping, we obtain the final representation of prescribed generic drug representation $e_{\widetilde{m}} = e_{\widetilde{f}^\circ} W_3 + b_3 \in \mathbb{R}^{1\times dim}$, where W_3 and b_3 are the weight matrix and bias, respectively.

We use $S\text{-}doctor$ to represent the entire Standardized doctor module, and the above process can be expressed as

$$e_{\widetilde{m}} = S\text{-}doctor(d, p) \tag{11}$$

3.5. Personalized patient information mining module

In this module, we capture the patient's personalized status and its changes, and combine it with the generic drug representation obtained in the $S\text{-}doctor$ module to achieve personalized drug recommendation for patients. Simultaneously, we obtain the specific drug representations of historical prescriptions through the preprocess module, thereby constraining and guiding the final personalized drug recommendations.

For the t-th historical visit of patient k, where $t \in \{1, \ldots, T_{k-1}\}$, the corresponding diagnosis record set is $d_t^k = \{d_{t,1}^k, \ldots, d_{t,\Gamma_{t,d}^k}^k\}$, the procedure record set is $p_t^k = \{p_{t,1}^k, \ldots, p_{t,\Gamma_{t,p}^k}^k\}$, and the drug record set is $m_t^k = \{m_{t,1}^k, \ldots, m_{t,\Gamma_{t,m}^k}^k\}$. For the current visit T_k of patient k, the corresponding diagnosis and procedure record sets are $d_{T_k}^k$ and $p_{T_k}^k$, respectively. The objective of the drug recommendation module is to predict the multi-hot vector $y^k \in [0,1]^{|M|}$ corresponding to the ground-truth drug record set $m_{T_k}^k$.

First, we use $S\text{-}doctor$ to obtain the generic predicted drug representations $e_{\widetilde{m}_t^k}$ for patient k's historical visits and $e_{\widetilde{m}_{T_k}^k}$ for current visit T_k, respectively.

$$\begin{aligned} e_{\widetilde{m}_t^k} &= S\text{-}doctor(d_t^k, p_t^k) \\ e_{\widetilde{m}_{T_k}^k} &= S\text{-}doctor(d_{T_k}^k, p_{T_k}^k) \end{aligned} \tag{12}$$

Then, we use $pre\text{-}emb$ to transform the real historical drug records into corresponding drug representation $e_{m_t^k} \in \mathbb{R}^{1\times dim}$

$$e_{m_t^k} = \sum_{i=1}^{\Gamma_{t,m}^k} pre\text{-}emb(m_{t,i}^k) \tag{13}$$

The representation bias between generic and real drug representations is caused by the individualized patient status. We extract this difference and map it to the patient space to capture the status

$$e_{H_t}^k = (e_{m_t^k} - e_{\widetilde{m}_t^k}) W_H \tag{14}$$

of the patient k in t-th visit, where W_H is weight matrix for mapping the drug space to the patient space. Considering that the patient status changes with each visit, we employ the GRU to capture such dynamics to obtain the personalized status representation

$$e_{H_{T_k}}^k = GRU(e_{H_0}, e_{H_1}^k, \ldots, e_{H_{T_k-1}}^k) \tag{15}$$

of the patient for the current visit, where e_{H_0} is the globally shared patient status representation, representing the common attributes of all patients. Then, the predicted personalized drug representation $e_{m_{T_k}^k}^{(1)} \in \mathbb{R}^{1\times dim}$ of the patient is

$$e_{m_{T_k}^k}^{(1)} = e_{\widetilde{m}_{T_k}^k} + e_{H_{T_k}}^k W_M \tag{16}$$

where W_M is the weight matrix used to map the patient space to the drug space.

To further enhance the stability of the predicted patient-specific drug representations, we use historically prescribed real drugs as auxiliary information to guide the personalized drug representations. We calculate the attention $att_t^k \in \mathbb{R}^1$ between the real drugs prescribed in history and the generic drugs prescribed currently as

$$att_t^k = softmax(FFN(cat(e_{\widetilde{m}_{T_k}^k}, e_{m_t^k}))) \tag{17}$$

The attention is employed to enhance the importance of historical visits that exhibit high similarity with the current visit, while reducing the importance of historical visits with low similarity. Then, the real historical prescribed drugs are aggregated based on attention, resulting in the whole historical drug representation

$$e_{m_{1:T_k-1}^k} = \sum_{t=1}^{T_k-1} att_t^k e_{m_t^k} \tag{18}$$

where $e_{m_{1:T_k-1}^k} \in \mathbb{R}^{1\times dim}$.

Then, we employ cross-attention to facilitate the guidance of historical drugs to current drugs. We construct query matrix $Q_m \in \mathbb{R}^{1\times dim}$, key matrix $K_{\widetilde{m}} \in \mathbb{R}^{1\times dim}$ and value matrix $V_{\widetilde{m}} \in \mathbb{R}^{1\times dim}$ as

$$Q_m = e_{m_{1:T_k-1}^k} W_3^Q, K_{\widetilde{m}} = e_{\widetilde{m}_{T_k}^k} W_3^K, V_{\widetilde{m}} = e_{\widetilde{m}_{T_k}^k} W_3^V \tag{19}$$

where W_3^Q, W_3^K and W_3^V are weight matrices. Similar to Eq. (9) and Eq. (10), the patient-specific drug representation $e^{(2)}_{m_{T_k}^k} \in \mathbb{R}^{1\times dim}$ for auxiliary prediction is calculated.

Then the final drug representation $e^{rec}_{m_{T_k}^k} \in \mathbb{R}^{1\times dim}$ for recommendation is

$$e^{rec}_{m_{T_k}^k} = \sigma(\lambda) e^{(1)}_{m_{T_k}^k} + (1-\sigma(\lambda)) e^{(2)}_{m_{T_k}^k} \tag{20}$$

where λ is a trainable parameter, and σ is the sigmoid activation function, which is used to constrain the coefficient to the interval $[0, 1]$. Constructing a predictor prediction composed of multiple layers of fully connected networks, the final prediction probability $\widetilde{y} \in \mathbb{R}^{1\times|M|}$ of each drug prescribed to patient k can be obtained as

$$\widetilde{y}^k = prediction(e^{rec}_{m_{T_k}^k}) \tag{21}$$

3.6. Model training

In this module, we construct the loss function to jointly optimize the model. First, we minimize the DDI loss

$$L_{ddi}^k = sum((\widetilde{y}^k)^{Tr} \widetilde{y}^k \odot A) \tag{22}$$

to enhance the safety of drug interactions, where $\odot$ denotes element-wise multiplication. In general, too small DDI loss may affect the performance of drug recommendation. Therefore, to balance safety and accuracy, similar to previous methods (Li et al., 2024b; Mu et al., 2025; Yang et al., 2021b, 2023b), we introduce a threshold ϕ and set it to 0.06 by default. When the calculated DDI rate ρ is higher than ϕ, the model incorporates the L_{ddi} loss, otherwise it dose not.

Then, considering that the patient's status has a certain degree of stability, we use the loss constraint L_{simi}^k to maintain a high similarity of the patient's status across different visits as

$$L_{simi}^k = max(0, 1 - \frac{2}{T_k(T_k-1)} \sum_{\substack{e_i, e_j \in L \\ e_i \neq e_j}} cosine(e_i, e_j)) \tag{23}$$

where $L = \{e_{H_0}, e^k_{H_1}, \ldots, e^k_{H_{T_k-1}}\}$, $cosine$ is cosine similarity.

For the loss function of drug predictione, we use

$$L_{bce}^k = -\sum_{i=1}^{|M|} y_i^k \log(\widetilde{y}_i^k) + (1-y_i^k)\log(1-\widetilde{y}_i^k) \tag{24}$$

to make the predicted probabilities approach the real probabilities, and use the loss

$$L_{multi}^k = \frac{1}{|M|} \sum_{i,j: y_i^k=1, y_j^k=0} \max(0, 1 - (\widetilde{y}_i^k - \widetilde{y}_j^k)) \tag{25}$$

to maximize the difference in predicted recommendation probabilities between the real prescribed drugs and the remaining drugs.

Table 1
Statistics of the datasets.

Items	MIMIC-III	MIMIC-IV
# of patients	6350	60,125
# of visits	15,032	156,810
# of diagnoses	1958	2000
# of procedures	1430	1500
# of drugs	131	131
avg # of visits	2.37	2.61
avg # of drugs	11.44	6.66

Therefore, the final loss function of PAUP is

$$L_{total} = \sum_{k=1}^{N} L_{bce}^{k} + \alpha L_{multi}^{k} + \beta L_{simi}^{k} + \omega L_{ddi}^{k} \tag{26}$$

where α and β are hyperparameters, $\omega = 0$ if $\rho < \phi$, else $\omega = \min(1, \exp(1 - \rho/\phi))$. These parameters are utilized to balance the influence of different losses on the recommendation results.

4. Experiment

In this section, we analyze the performance of PAUP in drug recommendation.

4.1. Datasets

We used two real-world datasets, MIMIC-III (Johnson et al., 2016) and MIMIC-IV (Johnson et al., 2023), for experiments. MIMIC is a large publicly available intensive care unit (ICU) dataset comprising de-identified health-related data from patients at the Beth Israel Deaconess Medical Center in the United States. MIMIC-III encompasses data from 2001 to 2012, while MIMIC-IV includes data from 2008 to 2018. The statistical information of the datasets is shown in Table 1.

4.2. Baselines and evaluation indicators

To validate the performance of our model, we used 11 baselines for comparison, which are LR (Logistic Regression), ECC (Read et al., 2011), RETAIN (Choi et al., 2016), LEAP (Zhang et al., 2017), GAMENet (Shang et al., 2019b), SafeDrug (Yang et al., 2021b), MICRON (Yang et al., 2021a), COGNet (Wu et al., 2022), MoleRec (Yang et al., 2023b), CausalMed (Li et al., 2024b), and BiMoRec (Mu et al., 2025). LR is a simple linear classifier, ECC concatenates multiple classifiers to enhance the performance of multi-label classification, RETAIN and MICRON employ attention and recurrent residual networks, respectively, as the backbone of the model, LEAP decomposes treatment advises into a sequence of continuous decision processes. GAMENet is the first to consider both patient history and drug-drug interactions, while SafeDrug and MoleRec introduce molecular structural information of drugs. COGNet uses a copy-or-predict mechanism which frames drug recommendation as a sequence generation problem, CausalMed leverages causal explanations for dynamic differences under various health status, and BiMoRec incorporates the 3D geometric structure of drugs to enhance drug representations. Both the baselines and our proposed PAUP were implemented based on the PyHealth (Yang et al., 2023a) package.

We employed four metrics to evaluate the performance of the model: Jaccard Similarity Score (Jaccard), Precision Recall AUC (PRAUC), Average F1 score (F1-score), and Drug-Drug Interaction rate (DDI rate). A lower DDI rate indicates higher drug safety, while higher values in other metrics denote greater recommendation accuracy. Additionally, we provided the average number of drugs recommended by each model.

4.3. Parameter settings

For a fair comparison, we adopted the same parameter settings as the main baseline algorithm BiMoRec as possible. Specifically, we used the same dataset splits for training, validation, and testing, set the learning rate to $5e-4$, Adam to the optimizer, dropout rate to 0.5, DDI (Drug-Drug Interaction) threshold to 0.06, and the number of epochs to 20. For the hyperparameters α and β proposed in the paper, we aimed to control the scale of different loss values to be in the same order of magnitude. To achieve this, considering that L_{multi} involves pairwise loss calculations, we performed a grid search with $\alpha \in \{0.1, \ldots, 1\}$ and $\beta \in \{0.01, \ldots, 0.1\}$ to determine the optimal hyperparameter settings for the best recommendation performance of the model.

4.4. Recommendation performance evaluation

In this section, we compared the recommendation performance of PAUP and baseline algorithms across different metrics. Since we used parameter settings identical to those of the baseline algorithms, except for the hyperparameters, we referenced the experimental results reported in the baselines (Li et al., 2024b; Mu et al., 2025) to preserve their optimal recommendation performance

Table 2
The comparison of recommendation performance.

Model	MIMIC-III					MIMIC-IV				
	DDI↓	Jaccard↑	F1↑	PRAUC↑	Avg.#Med	DDI↓	Jaccard↑	F1↑	PRAUC↑	Avg.#Med
LR	0.0830	0.4924	0.6490	0.7548	16.0489	0.0783	0.4569	0.6064	0.6613	8.5746
ECC	0.0817	0.4856	0.6438	0.7590	16.2578	0.0764	0.4327	0.6129	0.6530	8.7934
RETAIN	0.0879	0.4871	0.6473	0.7600	19.4222	0.0936	0.4234	0.5785	0.6801	10.9576
LEAP	0.0762	0.4526	0.6147	0.6555	18.6240	0.0688	0.4254	0.5794	0.6059	11.3606
GAMENet	0.0890	0.4994	0.6560	0.7656	27.7703	0.0898	0.4565	0.6103	0.6829	18.5895
SafeDrug	**0.0655**	0.5154	0.6722	0.7627	19.4111	**0.0604**	0.4487	0.6014	0.6948	13.6943
MICRON	0.0727	0.5219	0.6761	0.7489	19.2505	0.0691	0.4640	0.6167	0.6919	12.7701
COGNet	0.0839	0.5312	0.6744	0.7708	27.6335	0.0911	0.4775	0.6233	0.6524	18.7235
MoleRec	0.0726	0.5293	0.6834	0.7746	22.0125	0.0722	0.4744	0.6262	0.7124	13.4806
CausalMed	0.0709	0.5389	0.6916	0.7826	20.5419	0.0677	0.4899	0.6412	0.7338	14.4295
BiMoRec	0.0707	0.5422	0.6944	0.7858	20.3989	0.0696	0.4917	0.6425	0.7366	13.5291
PAUP	0.0736	**0.5454**	**0.6979**	**0.7896**	21.4357	0.0673	**0.4960**	**0.6473**	**0.7415**	13.2843

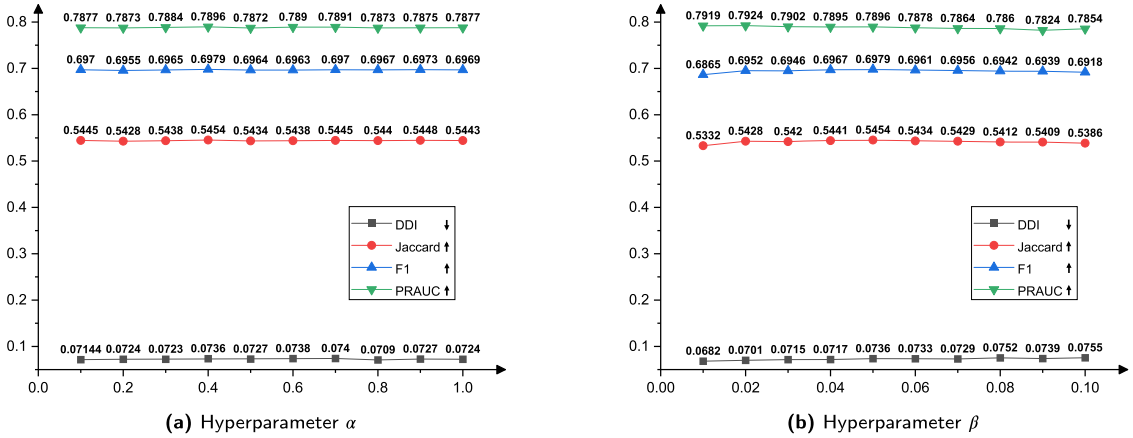


Fig. 2. Parameter sensibility analysis.

under hyperparameter tuning. The experimental results are presented in Table 2, where the optimal performance for each metric is highlighted in bold, and the sub-optimal performance is underlined.

The experimental results show that PAUP achieved optimal recommendation performance across the Jaccard, F1 and PRAUC metrics, indicating that our proposed method significantly enhances the accuracy of drug recommendation. Moreover, PAUP achieved competitive results on the DDI metric, suggesting that the method effectively ensures the safety of the recommended drugs. In summary, PAUP is capable of accurately and reasonably recommending safe drug combinations tailored to the real status of patients. SafeDrug achieved the best DDI rate on both datasets, highlighting its advantage in drug safety. However, its simplistic modeling of diagnosis data negatively impacts the accuracy of drug recommendations. BiMoRec achieved the second-best overall performance, demonstrating that incorporating the 3D structure of drugs can effectively improve recommendation performance. Inspired by this approach, our proposed PAUP also introduced drug 3D structural information.

4.5. Parameter sensibility analysis

In this section, we conducted a brief sensitivity analysis of the key hyperparameters α and β of the model on the MIMIC III dataset. The experimental results are presented in Fig. 2. The model's performance remains generally stable across different values of α and β, indicating that the model is not sensitive to these hyperparameters. Furthermore, compared to the baseline, the model achieved optimal performance in terms of Jaccard, F1 and PRAUC metrics across all hyperparameter settings. This demonstrates that the proposed PAUP significantly improves the performance of drug recommendation, rather than merely relying on hyperparameter tuning. These results validate the effectiveness and reliability of PAUP.

4.6. Ablation study

In this section, we performed ablation experiments to investigate the impact of each component on the model's performance. The experimental results are shown in Table 3, where 'w/o preprocess' refers to a model without the drug representation preprocess module, 'w/o status' refers to a model without the personalized user status-based drug representation, and 'w/o guide' refers to a model without the historical drug-guided representation. The superior performance of PAUP compared to the 'w/o preprocess' model

Table 3
Ablation study.

Methods	DDI↓	Jaccard↑	F1↑	PRAUC↑
w/o preprocess	0.0742	0.5433	0.6934	0.7875
w/o status	0.0713	0.5419	0.6947	0.7833
w/o guide	0.0720	0.5372	0.6907	0.7838
PAUP	0.0736	**0.5454**	**0.6979**	**0.7896**
PAUP - DDI	**0.0711**	0.5437	0.6965	0.7887

Table 4
Performance comparison of model accuracy.

Methods	Jaccard↑	F1↑	PRAUC↑
LR	0.4924	0.6490	0.7548
RETAIN	0.4871	0.6473	0.7600
LEAP	0.4526	0.6147	0.6555
GAMENet	0.5013	0.6574	0.7685
SafeDrug	0.5252	0.6800	0.7773
MICRON	0.5180	0.6729	0.7519
MoleRec	0.5325	0.6865	0.7801
CausalMed	0.5434	0.6959	0.7903
BiMoRec	0.5410	0.6940	0.7895
PAUP	**0.5499**	**0.7014**	**0.7954**

indicates that the drug representations obtained through preprocessing effectively capture the physicochemical properties of drugs. In comparison to PAUP, the accuracy of 'w/o status' significantly decreases across all metrics. We believe that this decline may stem from the absence of personalized user status constraints, leading to the bias of status information from other patients in general drug representation. In addition, the experimental outcomes of 'w/o guide' show the facilitating role of historically prescribed drugs in current diagnosis. The DDI rate under 'w/o status' and 'w/o guide' is better than that of PAUP. We speculate that this is because richer drug representations in PAUP may limit the model's ability to enforce DDI constraints. This suggests that, compared to single-sided drug representations ('w/o status' or 'w/o guide'), PAUP, which employs bilateral drug representations, requires stronger DDI loss constraints under otherwise identical conditions.

Therefore, we incorporated the experimental results of PAUP under a stricter DDI constraint (DDI = 0.0711) as 'PAUP-DDI'. The experimental results of 'PAUP-DDI' indicate that PAUP can achieve significantly superior drug recommendation performance compared to the ablation models under stricter DDI constraints, highlighting the necessity of all three components.

4.7. In-depth analysis

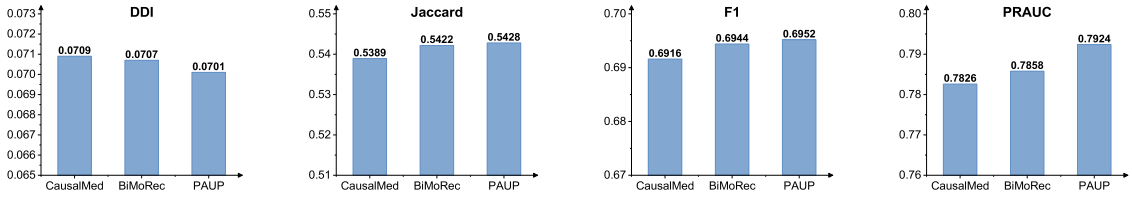
4.7.1. Performance comparison of model accuracy

Drug recommendation exhibits a certain tolerance towards the risks of DDI, which primarily focus on the accuracy of the recommended drug combinations. Therefore, in this subsection, we eliminated the constraints on the DDI rate both in the baseline models and in our proposed PAUP model, focusing solely on comparing the accuracy of drug combination across different models. The experimental results are presented in Table 4, where BiMoRec[2], CausalMed[3], MoleRec[4], MICRON[5], and SafeDrug[6] are implemented using their respective official codes, and the implementation of the GAMENet model is based on the official code provided by MICRON. The LR, RETAIN and LEAP models did not impose constraints on the DDI rate. Therefore, we retained their experimental results. The results indicate that PAUP achieves the optimal performance across all three evaluation metrics, which shows the model's effectiveness in improving the accuracy of drug recommendations.

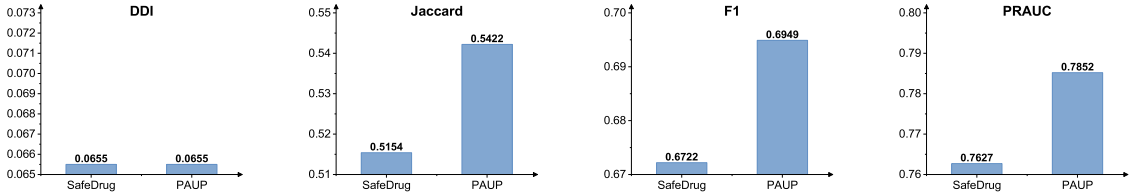
4.7.2. Performance comparison under the similar drug safety

In this subsection, we evaluated the difference in accuracy between PAUP and the two primary baseline algorithms, CausalMed (DDI rate = 0.0709) and BiMoRec (DDI rate = 0.0707), under the condition of similar drug safety profiles (i.e., similar DDI rate). The experimental results are depicted in Fig. 3a, where PAUP exhibited a DDI rate of 0.0701. Furthermore, since the SafeDrug model achieved the optimal DDI rate (0.0655), we also included this model for comparison. The experimental results are illustrated in Fig. 3b, where PAUP exhibited a DDI rate of 0.0655. From Fig. 3a, it is evident that even with a superior DDI rate, PAUP maintained a higher accuracy, indicating a performance advantage of our proposed model in the field of drug recommendation. Fig. 3b reveals

[2] https://github.com/guangyunms/BiMoRec
[3] https://github.com/lixiang-222/CausalMed
[4] https://github.com/yangnianzu0515/MoleRec
[5] https://github.com/ycq091044/MICRON
[6] https://github.com/ycq091044/SafeDrug



(a) Comparison with primary baselines



(b) Comparison with the model of optimal safety

Fig. 3. Performance comparison under the similar drug safety.

Table 5
Comparison of training and inference times.

Methods	Convergence epoch	Training time/ Epoch(s)	Total training Time(s)	Inference Time(s)
MoleRec	44	365	16,060	15.2
CausalMed	48	676	32,448	75.9
BiMoRec	19	462	8778	15.7
Preprocess	66	0.45	29.7	-
PAUP	5	190	950	9.8

that under the same DDI rate, PAUP's recommendation accuracy significantly surpassed that of SafeDrug, demonstrating PAUP's capability to enhance recommendation accuracy while simultaneously considering the safety of drug combinations.

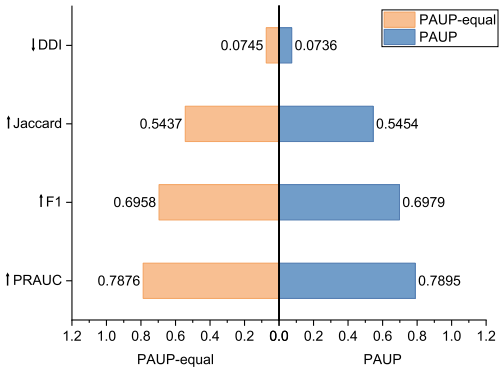
4.7.3. Performance comparison of differences in patient status handling

In this subsection, we conducted a further analysis of patient status. The experimental results are shown in Fig. 4, where 'PAUP-equal' represents identical patient status representations across historical visits, 'PAUP-diff' represents patient status without similarity constraints, 'PAUP-mean' is the mean representations of historical visits, and 'PAUP-att' aggregates representations of historical visits through attention-weighted summation. The results in Fig. 4a show that patient status is not fixed, as there are status differences across historical visits. The results in Fig. 4b indicate that the changes in patient status are gradual, showing a certain degree of similarity across different visits. The results in Fig. 4c show that simply comparing the current patient status with the overall historical mean status is insufficient. The results in Fig. 4d indicate that the sequential information of patients' historical visits contributes positively to the model's performance. Furthermore, PAUP achieved the optimal drug recommendation performance across all evaluation metrics, indicating that the model effectively captures the dynamic changes in patient status.

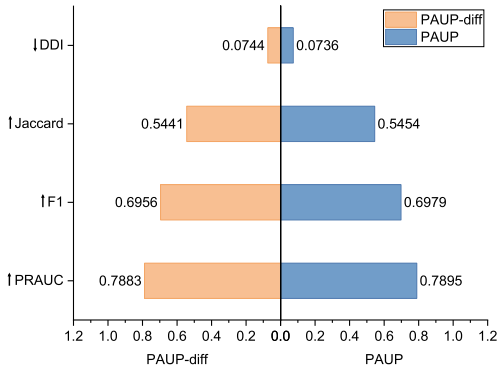
4.7.4. Time complexity analysis

In this subsection, we provide a brief analysis of the time complexity of PAUP. Specifically, (a) in the preprocessing part, the time complexity of GIN is $\mathcal{O}((G_n d^2 + G_e d)l)$, where G_n and G_e represent the total number of nodes and edges in the molecular structure and substructure respectively, d is the embedding dimension, and l is the number of network layers. The time complexity of GVP is $\mathcal{O}(G_n(s_{in}s_{out} + s_{in}v_{out}v_{in} + v_{out}v_{in}k))$ (Mu et al., 2025), where s and k represent the dimension of scalar and vector features, and v represents the number of channels for the vector. The time complexity of cross-attention is $\mathcal{O}(G_n d)$, where we remove the scalar sequence length with a value of 2. The time complexity of the loss function is $\mathcal{O}((B_1^2 + B_2^2 + B_3^2)d)$, where B_1, B_2 and B_3 denote the sampling numbers for molecules, substructures, and drugs. In summary, the time complexity of this part is linearly correlated with $\mathcal{O}(max(G_n, G_e))$. (b) In the model training part, the time complexity of the Standardized Doctor module is $\mathcal{O}((\Gamma_d + \Gamma_p) \cdot d + d^2)$, where Γ_d and Γ_p represent the number of diagnosis and procedure records accessed in a single visit. The time complexity of the Drug Recommendation Module is $\mathcal{O}((((\Gamma_d + \Gamma_p) \cdot d + d^2)L_s + d^2 + d + M^2 + Md)N)$, where N is the number of patients, L_s is the number of visits for a single patient, and M is the number of drugs. The time complexity of the DDI module is $\mathcal{O}(NM^2\, d)$. In conclusion, the time complexity of this part is linearly correlated with $\mathcal{O}(N)$. Therefore, the overall time complexity of PAUP is $\mathcal{O}(N)$, which aligns with the main baseline algorithms (MoleRec, CausalMed, BiMoRec).

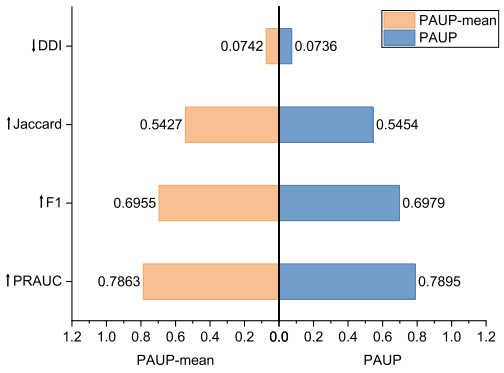
Furthermore, we compared the actual runtimes of PAUP and the main baseline algorithms in the same environment (RTX4090 GPU, Pytorch 1.13.0, CUDA 11.7). The experimental results are shown in Table 5. The experimental results show that the proposed



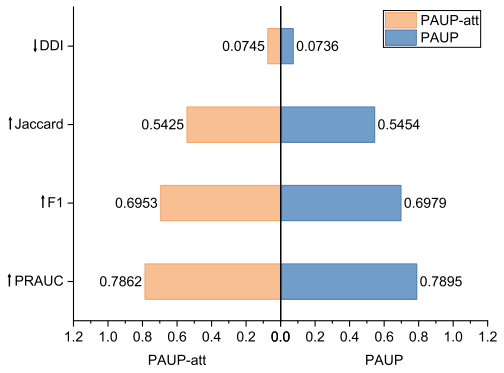
(a) Comparison with identical patient status across historical visits



(b) Comparison with patient status without similarity constraints



(c) Comparison with mean representations of historical visits



(d) Comparison with representations through attention-weighted summation

Fig. 4. Performance comparison of differences in patient status handling.

PAUP model achieved faster convergence, and realized less time for both training and inference. This indicates that, compared to the main baselines, PAUP exhibits a significant advantage in efficiency.

4.7.5. Case study

In this section, as illustrated in Table 6, we utilize specific cases to show the generic drug set prescribed after obtaining current clinical records and the patient-specific drug set prescribed after obtaining patient-specific information. Then, we attempt to provide a brief analysis of generic drugs and patient-specific drugs. To simplify the description, we omitted consistent drugs (such as Stomatological preparations (A01A), Other beta-lactam antibacterials (J01D), and Laxatives (A06A), etc.), as well as some non-essential symptomatic drugs (such as Antiemetics and antinauseants (A04A), Opioids (N02A), and Antiepileptics (N03A), etc.). The case analysis is as follows.

Beta-lactam antibacterials, Penicillins (J01C) in generic drugs possess anti-infective properties. Considering the patient's history of MRSA septicemia (038.12), there may be a risk of colonization or infection with methicillin-resistant staphylococcus aureus. This could render Penicillin-class drugs ineffective against MRSA, potentially leading to a delay in the appropriate treatment. Therefore, the J01C was removed. In the event of infection in this patient, Corticosteroids for Systemic Use, Plain (H02A) is incorporated into the specific patient's treatment drugs to manage the infection and inflammatory reaction (996.62) and other associated symptoms. In addition, the patient has a long-standing severe renal condition and medical history of Kidney dialysis (E879.1). Therefore, although High-Ceiling Diuretics (C03C) can be used in treating Chronic diastolic heart failure (428.32), given a comprehensive evaluation, the drug is removed in patient-specific drugs. The patient-specific drugs add All Other Therapeutic Products (V03A). The category is quite diverse. Considering the patient's history of long-term blood transfusions, this situation may lead to iron overload. Therefore, we speculate that the possible reason is the use of Iron chelating agents (V03AC). Furthermore, some Lipid modifying agents, plain (C10A), such as Bempedoic acid (C10AX15), may potentially trigger Gout (274.9). We speculate that this is the reason for its removal.

Table 6
Case study.

Visit 1	diagnoses	038.12, 053.19, 244.9, 274.9, 276.7, 286.9, 287.5, 403.91, 416.8, 427.31, 427.81, 428.0, 428.33, 511.9, 585.6, 707.03, 707.21, 785.52, 995.92, 996.62, E879.1, V45.01
	procedures	38.91, 38.93, 38.95, 39.95, 86.05
Visit 2	diagnoses	285.9, 287.5, 403.90, 427.31, 428.0, 428.33, 584.9, 585.4, 599.0, V45.01, V58.61
	procedures	38.95, 39.95
Visit 3	diagnoses	008.45, 038.9, 198.5, 244.9, 285.9, 287.5, 349.82, 403.91, 416.8, 427.31, 428.0, 428.32, 486, 518.81, 564.00, 585.6, 707.03, 707.22, 785.52, 933.1, 995.92, E849.7, E912, V10.3, V45.01, V45.11, V49.86, V58.61
	procedures	38.91, 38.95, 38.97, 96.04, 96.6, 96.71
Medications	generic	A01A, A02B, A06A, A12A, A12C, A07A, A12B, B05C, B01A, B02B, C01C, C01B, H03A, J01D, N02B, N06A, N05C, N05A, R03A, A04A, C07A, C03C, C10A, J01C, N01A, N02A
	specific	A01A, A02B, A06A, A12A, A12C, A07A, A12B, B05C, B01A, B02B, C01C, C01B, H03A, J01D, N02B, N06A, N05C, N05A, R03A, H02A, N03A, V03A

5. Conclusion

In paper, we propose Robust Drug Recommendation Based on **P**atient Status **A**wareness and **U**nbiased **P**rediction (PAUP). This approach constructs a generic patient-agnostic drug prediction model to avoid prediction bias caused by differences among patients. Subsequently, for the first time, we incorporate personalized patient status information into the modeling process and combine it with the general prediction model to obtain drug prediction results tailored to specific patients. Additionally, we consider the differences in patient status across different visits, and track and capture these variations. We also calculate the similarity between historically prescribed drugs and the currently predicted drugs to guide drug recommendation more effectively. Moreover, we introduce 2D and 3D drug structural information to capture the physicochemical properties of drugs, thereby enhancing the performance of drug representation. In the future, we will further account for and remove the noise introduced by irrelevant historical visit records of target patient, and we will investigate denoising medical datasets to enhance dataset quality and drug recommendation performance. We plan to incorporate more comprehensive drug data to achieve fine-grained drug recommendations, including details such as drug dosage cycles, dosage amounts, and administration times. Furthermore, we aim to explore the use of large models for deeper drug data modeling.

CRediT authorship contribution statement

Gang-Feng Ma: Writing – original draft, Software, Methodology, Funding acquisition; **Xilin Wen:** Writing – review & editing, Validation; **Xu-Hua Yang:** Funding acquisition, Conceptualization; **Yanbo Zhou:** Project administration; **Wei Huang:** Visualization, Data curation; **Xiao-Xin Li:** Resources, Investigation; **Peng Jiang:** Supervision, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

None.

Acknowledgments

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